

OMY

The OMY hardware platform is a teleoperation system specifically designed for [imitation learning applications](#). It consists of two main components: the OMY-F3M follower robot arm and the OMY-L100 leader device. The follower robot features a [6-DOF](#) robot arm with high-precision DYNAMIXEL-Y actuators, capable of handling payloads up to [3kg](#) with sub-millimeter repeatability. The leader device provides intuitive 6-DOF control through a lightweight, ergonomic design that translates human hand movements into precise robot commands. This leader-follower system enables human operators to demonstrate complex manipulation tasks that can be learned and replicated by AI systems, with real-time communication via Ethernet and high-resolution joint control for accurate data collection.

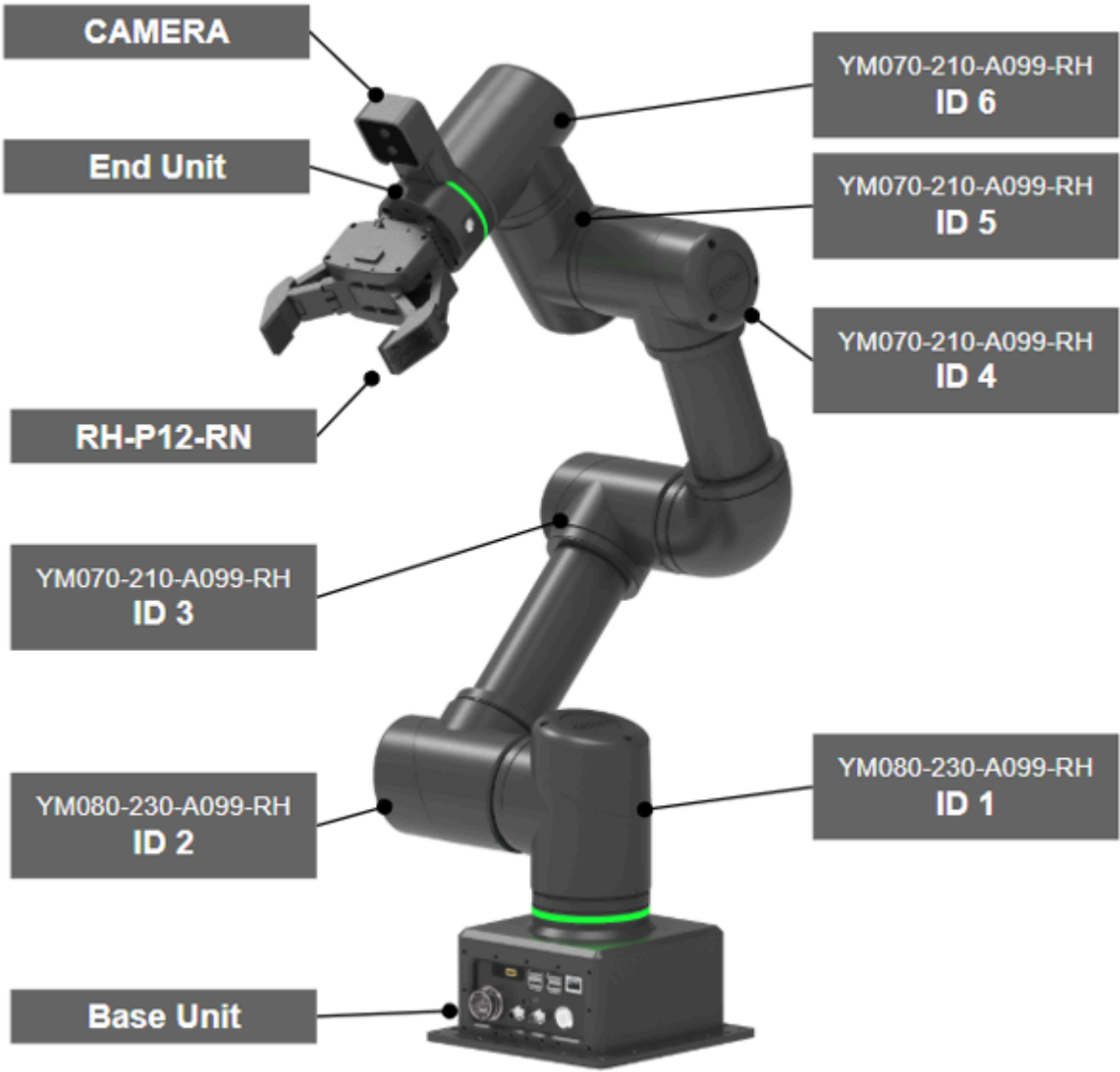


Included Common Items

Item	Qty
OMY-L100	1
OMY-F3M	1
USB Hub 3.0	1
USB cable	1
SMPS for OMY-F3M	1
SMPS for OMY-L100	1

Item	Qty
E stop	1

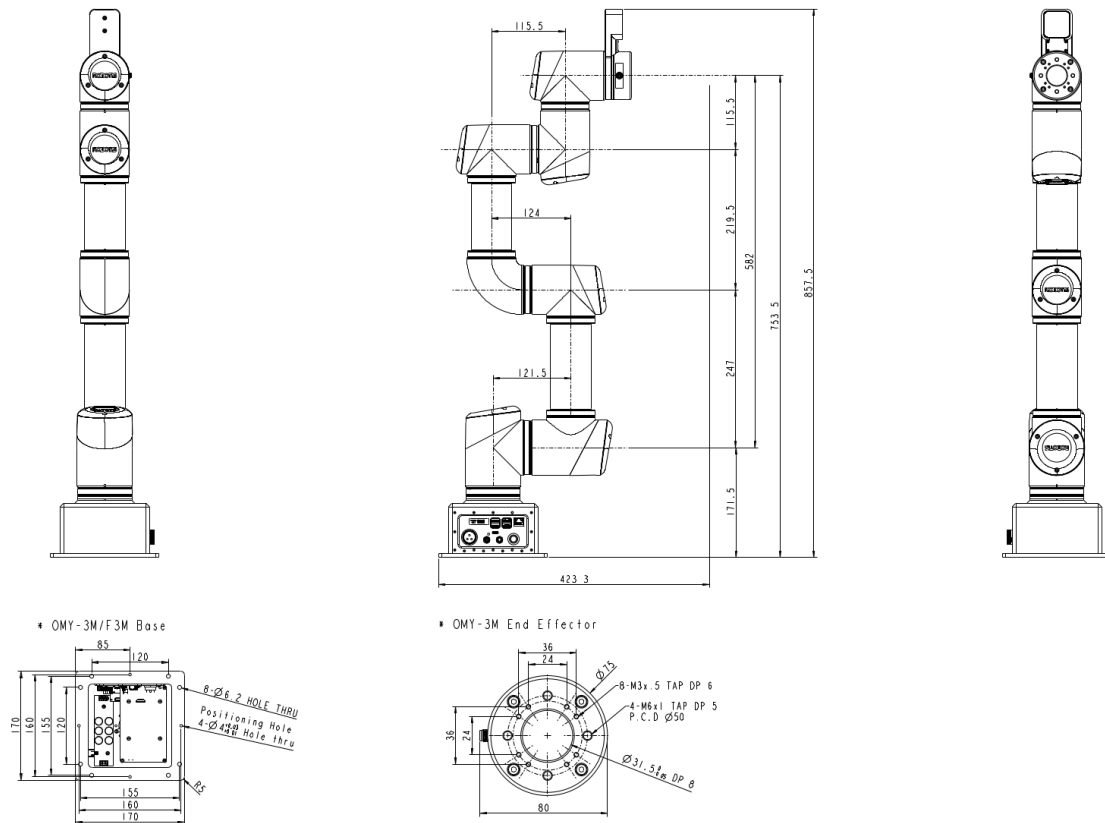
[Follower] Hardware Overview



Item	OMY-F3M
DOF	6
Payload	3 [kg]
Reach	580 [mm]
Weight	13.5 [kg]
Operating Voltage	24 [VDC]

Item	OMY-F3M
Joint Resolution	$-\pi(\text{rad}) \sim \pi(\text{rad})$, -262,144 ~ 262,144[pulse/rev]
Joint Range	Joint 1, 2 : $\pm 360^\circ$ Joint 3 : $\pm 150^\circ$ Joint 4, 5, 6 : $\pm 360^\circ$
DYNAMIXEL-Y Specification	Joint 1,2 : YM080-230-A099-RH Joint 3,4,5,6 : YM070-210-A099-RH
Repeatability	± 0.05 [mm]
Brake Force	200% of the continuous torque for each joint
TCP Speed	< 900 [mm/s]
Host Interface	Ethernet
Internal Communications	RS485
Communication Baudrate	4 [Mbps]
Robot Hand	RH-P12-RN
Camera	Intel RealSense D405

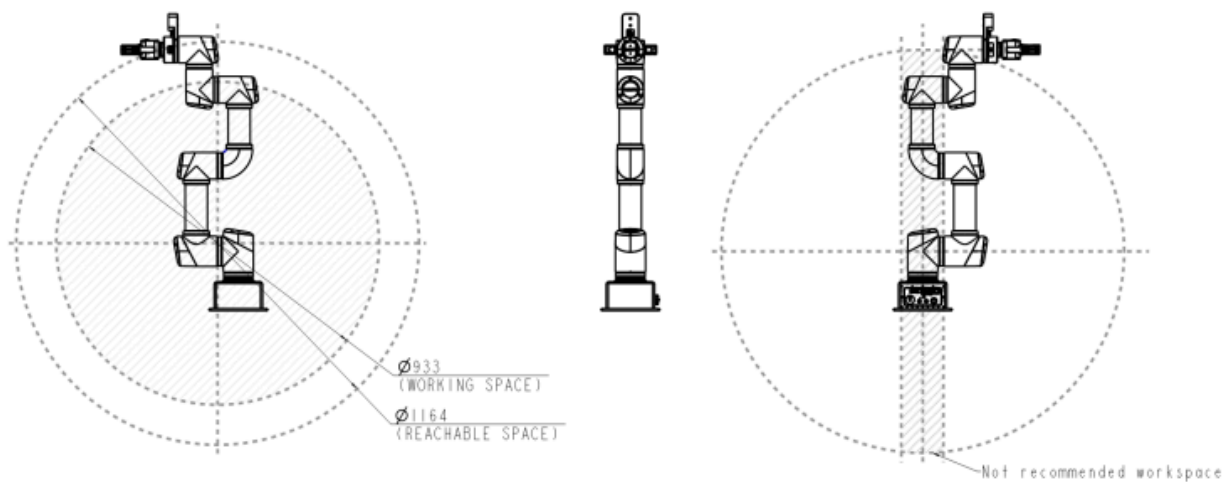
[Follower] Layout



Drawing Files

File Type	Download Link
PDF	Download PDF
DWG	Download DWG
STEP	Download STEP

[Follower] Workspace



Drawing File

File Type	Download Link
PDF	 Download PDF

[Follower] Inertia

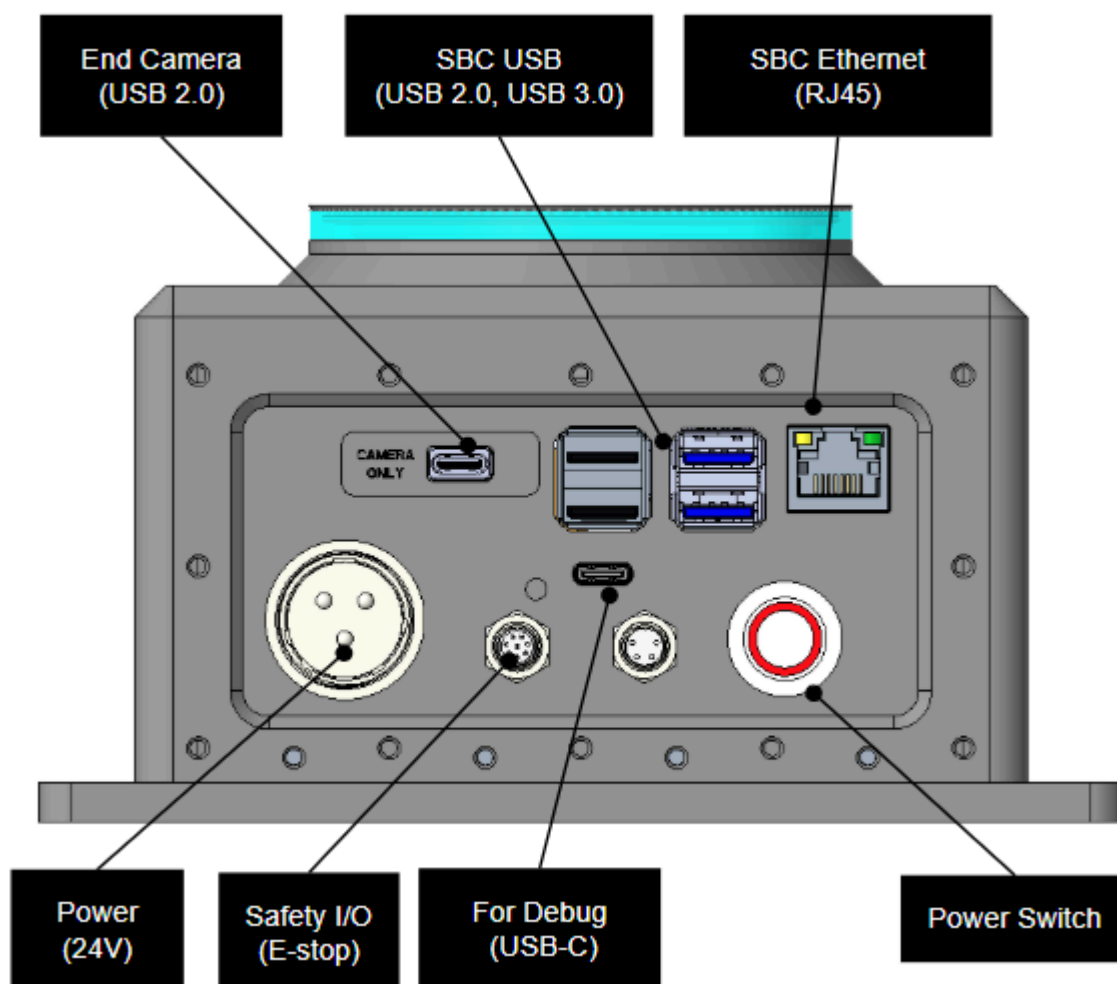
Joint	Mass [g]	COG X [mm]	COG Y [mm]	COG Z [mm]	Inertia [kg·m²]
1	2.0648832e+03	-1.1063615e-01	-5.4711270e+00	-1.5897733e+01	3.033
2	3.6795395e+03	1.1405379e-02	1.6184244e+01	1.0360634e+02	5.173
3	2.3865916e+03	7.8148689e-02	1.0718481e+02	1.4117267e+02	2.2694
4	1.4002347e+03	-1.3318256e-01	-1.0746667e+02	1.9090688e+01	1.7541
5	1.4002347e+03	1.3318256e-01	-1.9090688e+01	1.0996667e+02	1.7541

Joint	Mass [g]	COG X [mm]	COG Y [mm]	COG Z [mm]	I _z
6	4.0015475e+02	1.1080146e-02	-8.0594056e+01	9.3029337e+00	5.674

► Raw inertia data (full)

[Follower] Base Unit

Back Panel



Pin Assignment

Power Connector(Panel)

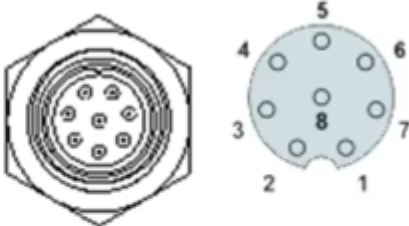


① 24V

② GND

③ FG

Safety I/O (E-stop)



① EM1

② COM

③ EM2

④ COM

⑤ NC

⑥ NC

⑦ NC

⑧ NC

[Follower] End Unit

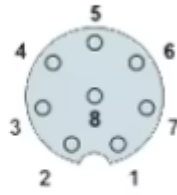
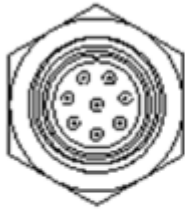


End Tool Support Specifications

Item	Specification
Power Output	24V / 1A
Interface	RS485, Max 4 Mbps
Protocol	Dynamixel Protocol 2.0, Modbus-RTU(Only 30001~39999, 40001 ~ 49999)
Connector	M8 8pin

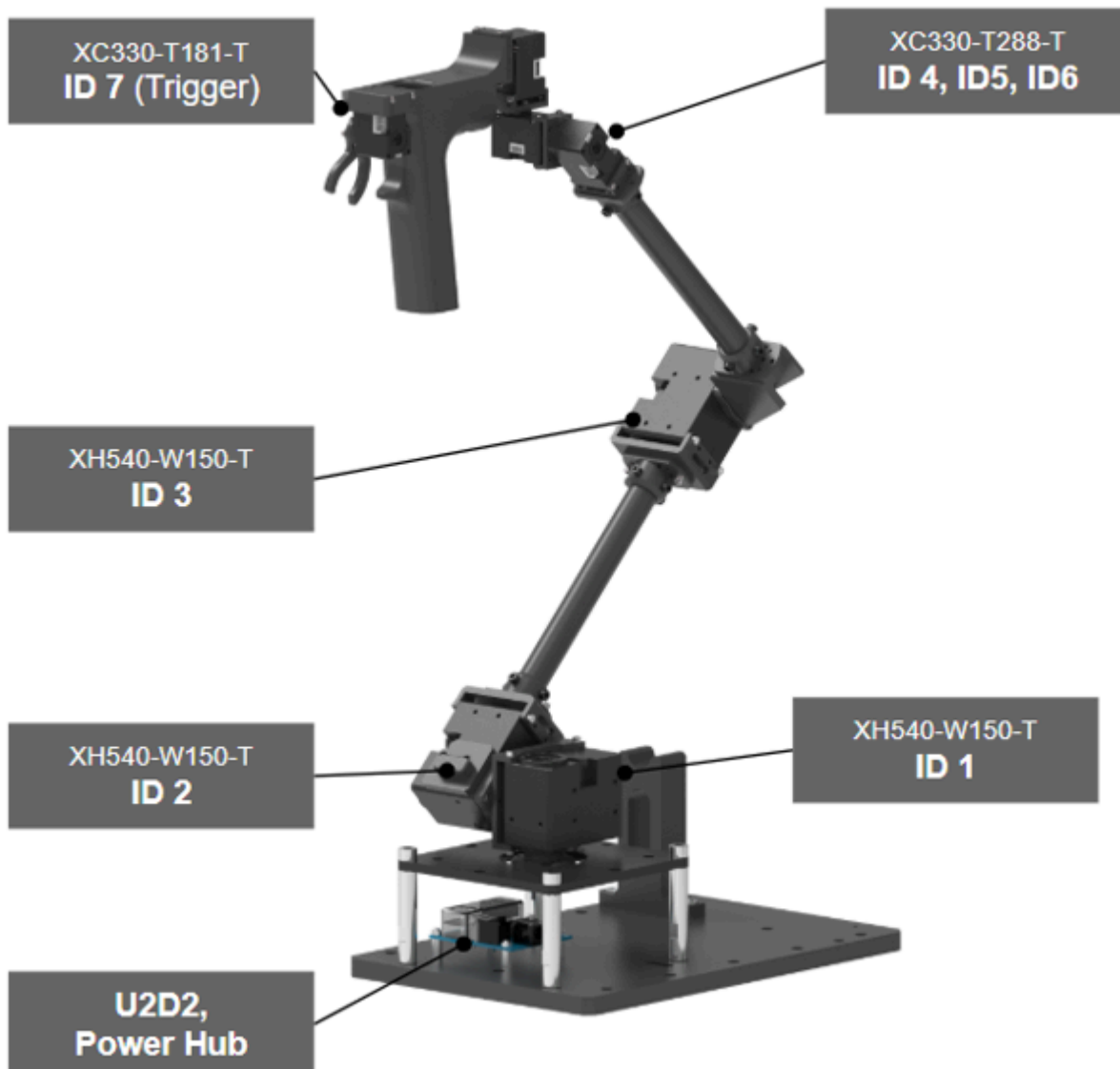
Pin Assignment

End I/O (M8 Connector)



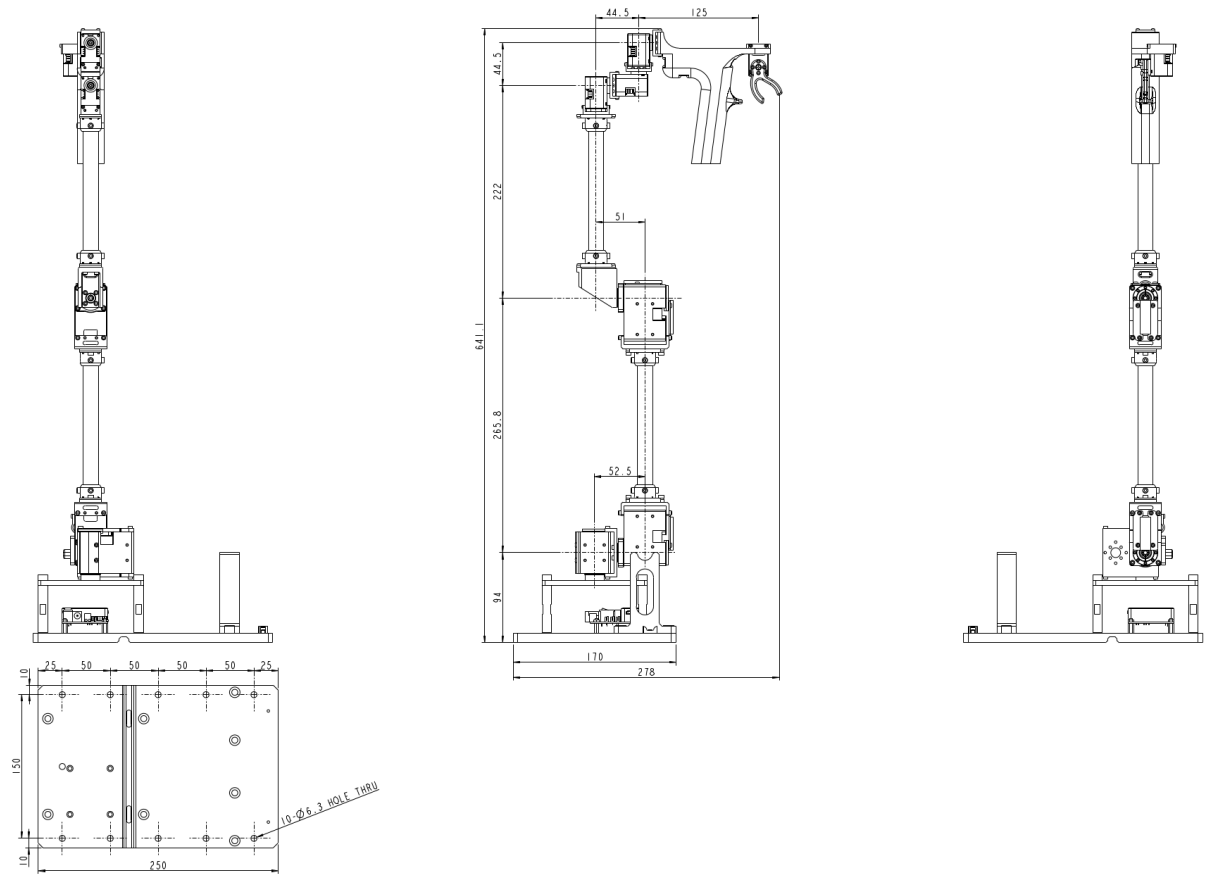
- | | |
|----------------|-----------|
| ① AI1/RS485 DN | ⑤ PWR 24V |
| ② AI0/RS485 DP | ⑥ DI0 |
| ③ DO0 | ⑦ DI1 |
| ④ DO1 | ⑧ PWR 0V |

[Leader] Hardware Overview



Item	OMY-L100
DOF	6
Reach	560 [mm]
Weight	1.46 [kg]
Operating Voltage	12 [VDC]
Joint Resolution	$-\pi(\text{rad}) \sim \pi(\text{rad})$, -2,048 ~ 2,048 [pulse/rev]
Joint Range	Joint 1 : $\pm 180^\circ$ Joint 2 : $-70^\circ \sim +100^\circ$ Joint 3, 4, 5, 6 : $\pm 180^\circ$ Joint 7 : $-90^\circ \sim +60^\circ$
DYNAMIXEL-Y Specification	Joint 1, 2, 3 : XH540-W150 Joint 4,5,6 : XC330-T288 Joint 7 : XC330-T181
Host Interface	U2D2 (USB 2.0)
Internal Communications	TTL Multidrop Bus
Communication Baudrate	4 [Mbps]

[Leader] Layout



Drawing Files

File Type	Download Link
PDF	 Download PDF
DWG	 Download DWG
STEP	 Download STEP

[Leader] Inertia

Joint	Mass [g]	COG X [mm]	COG Y [mm]	COG Z [mm]	
1	1.9792620e+02	-1.5094777e+01	-2.9152531e+00	-5.8029327e-01	6.2
2	4.9547242e+02	-6.6509235e-02	2.2442703e+00	1.3349620e+02	6.

Joint	Mass [g]	COG X [mm]	COG Y [mm]	COG Z [mm]	
3	9.8660130e+01	-1.7428771e-01	4.9723865e+01	1.2400449e+02	6
4	2.6687200e+01	2.6921263e-02	-3.5455461e+01	7.5765923e-01	4.9
5	2.6687200e+01	-2.6921263e-02	-7.5765923e-01	3.3955461e+01	4.9
6	1.0912699e+02	3.7567212e+00	-7.7299588e+01	-3.2079617e+01	2.9
7	4.0609159e+00	-2.6354677e-01	-2.0547867e+00	-8.7853268e+00	7.9

► Raw inertia data (full)

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Control Table